Reinforcement Learning Control for 6 DOF Flight of Fixed-Wing Aircraft

固定翼飞机六自由度飞行的强化学习控制

ZHANG Sheng [1,2], DU Xin [1], XIAO Juan [1], HUANG Jiangtao [1], HE Kaifeng [1,2]

张胜 [1,2]，杜鑫 [1]，肖娟 [1]，黄江涛 [1]，何开峰 [1,2]

1. China Aerodynamics Research and Development Center, Mianyang 621000, China;

1. 中国空气动力研究与发展中心，绵阳 621000，中国；

2. State Key Laboratory of Aerodynamics, Mianyang 621000, China

2. 空气动力学国家重点实验室，绵阳 621000，中国

Abstract

摘要

Flight control is a key technique for the autonomous unmanned aircraft. The traditional model-based controller design approaches often aim at concrete plant and are short in universality. Reinforcement learning provides a general controller design paradigm that is adaptive, optimized, model-free and widely applicable, and it is a promising way for the intelligent control. In contrast to the 3 Degree-of-freedom (DOF) flight, the 6 DOF motion better describes the aircraft real flight, while the implementation of the intelligent control is much harder. Based on the multiple continuous states input and multiple continuous action output deep reinforcement learning, the integrated intelligent control for the cruise flight, directly from the vehicle flight states to the aero-surfaces and thrust control, is developed for the full-sized fixed-wing aircraft. It avoids the artificial trajectory and attitude loop separation in the traditional controller design, and is advantageous to the exploitation of the aerodynamics and the nonlinear inertia coupling.

飞行控制是无人机自主飞行的关键技术。传统的基于模型的控制器设计方法通常针对具体的飞行器，通用性较差。强化学习提供了一种自适应、优化、无模型且适用范围广的通用控制器设计范式，是实现智能控制的一种有前景的方法。与三自由度（3 DOF）飞行相比，六自由度（6 DOF）运动能更好地描述飞行器的真实飞行，但智能控制的实现难度也大得多。本文基于多连续状态输入和多连续动作输出的深度强化学习，为全尺寸固定翼飞机开发了一种巡航飞行的集成智能控制，直接从飞行器飞行状态到气动舵面和推力控制。它避免了传统控制器设计中人为的轨迹和姿态回路分离，有利于利用气动特性和非线性惯性耦合。

Key Words: fixed-wing aircraft; flight control; artificial intelligence; intelligent control; deep reinforcement learning

关键词：固定翼飞机；飞行控制；人工智能；智能控制；深度强化学习

1 INTRODUCTION

1 引言

Flight control is a key technique for the development of autonomous unmanned aircraft system. It includes the take-off and landing control, attitude hold, height control, and trajectory tracking. The PID control method is widely used for the aircraft flight control. However, due to the nonlinearity of the aircraft dynamics, the stability of the linear controller cannot be guaranteed theoretically. Moreover, the design procedure is time-consuming since it involves multiple trim points selection and control gain tuning [1]. The modern control technologies, including the LQR control [1], the dynamic inverse control [2], the backstepping control [3], the sliding mode control [4], and the L1 adaptive control [5], etc., have also been extensively studied for the aircraft flight control. In general, these control schemes achieve good performance. Nevertheless, the PID control and modern control methods are model-based. In practice, there always exist errors and disturbances for the plant modeling. For the complex system like the aircraft, which contains mechanics, electricity, and structure sub-systems, although various robust control approaches are proposed, the potential poorly-understood uncertainties, such as the nonlinear unsteady aerodynamics at high angle-of-attack [6], may ruin the control effect or even bring disaster.

飞行控制是自主无人机系统发展的关键技术。它包括起降控制、姿态保持、高度控制和轨迹跟踪。PID 控制方法被广泛应用于飞行器飞行控制。然而，由于飞行器动力学的非线性，线性控制器的稳定性在理论上无法得到保证。此外，设计过程耗时，因为它涉及多个配平点选择和控制增益调整[1]。现代控制技术，包括 LQR 控制[1]、动态逆控制[2]、反步控制[3]、滑模控制[4]和 L1 自适应控制[5]等，也已在飞行器飞行控制方面进行了广泛研究。总的来说，这些控制方案都取得了良好的性能。然而，PID 控制和现代控制方法都是基于模型的。在实践中，工厂建模总是存在误差和干扰。对于像飞机这样包含机械、电气和结构子系统的复杂系统，尽管提出了各种鲁棒控制方法，但潜在的、理解不足的不确定性，例如高攻角下的非线性非定常气动[6]，可能会破坏控制效果甚至带来灾难。

Artificial Intelligence (AI) may greatly develop the human productivity and bring historical changes into the human society. Learning is an essential feature of intelligence and currently Machine Learning (ML) is the most promising branch in the field of AI research, which may be classified as the supervised learning, unsupervised learning and reinforcement learning [7]. Among them, reinforcement learning is widely applicable for the dynamic

人工智能（AI）可能会极大地发展人类生产力，并给人类社会带来历史性变革。学习是智能的一个基本特征，目前机器学习（ML）是人工智能研究领域中最有前途的分支，可分为监督学习、无监督学习和强化学习[7]。其中，强化学习广泛适用于动态

decision-making problems. Based on such approach, agents can interact with the environment and learn behavioral strategies that maximize specific index [8]. After years of development, the value function-based reinforcement learning method and the direct policy search reinforcement learning method are proposed upon the theory of dynamic programming, and the Actor-Critic methods that integrate the two methods are formed [9-12]. Compared with the traditional control methods, reinforcement learning provides a general controller design paradigm that is adaptive, optimized, model-free, and widely applicable, and is a promising way to realize the intelligent control.

决策问题。基于这种方法，智能体可以与环境交互并学习最大化特定指标的行为策略[8]。经过多年的发展，基于价值函数的强化学习方法和直接策略搜索强化学习方法在动态规划理论的基础上被提出，并形成了将这两种方法结合起来的 Actor-Critic 方法[9-12]。与传统控制方法相比，强化学习提供了一种自适应、优化、无模型且适用范围广的通用控制器设计范式，是实现智能控制的一种有前途的方法。

Neural Network (NN) is an information processing system which imitates the structure and function of human brain. It can approximate arbitrary nonlinear function and is an effective modeling tool [13]. In particular, deep NN model can learn complex functions and produce better performance, through refining features among hidden layers [14]. Deep Reinforcement Learning (DRL) is the variation of reinforcement learning that employs the deep NN model. The Deep Q Network (DQN) algorithm is a well-known DRL method, which achieves the level of human players in Atari games. However, it can only solve problems with “discrete action” [15]. To address such issue, the Deep Deterministic Policy Gradient (DDPG) algorithm is further developed for the robot motion control [16], and is now a popular algorithm for the “continuous state, continuous action” problems. With the DDPG algorithm, the intelligent control of the quadrotor hover and forward flight is realized in Ref. [17].

神经网络（NN）是一种模仿人脑结构和功能的信息处理系统。它能够逼近任意非线性函数，是一种有效的建模工具[13]。特别是，深度神经网络模型能够通过在隐藏层之间提炼特征来学习复杂函数并产生更好的性能[14]。深度强化学习（DRL）是强化学习的一种变体，它采用了深度神经网络模型。深度 Q 网络（DQN）算法是一种著名的 DRL 方法，它在 Atari 游戏中达到了人类玩家的水平。然而，它只能解决“离散动作”问题[15]。为了解决这个问题，深度确定性策略梯度（DDPG）算法被进一步开发用于机器人运动控制[16]，现在是解决“连续状态、连续动作”问题的流行算法。通过 DDPG 算法，文献[17]实现了四旋翼无人机悬停和前飞的智能控制。

On the other hand, the control of the aircraft flight often separate the plant into the outer-loop trajectory control and the inner-loop attitude control [18]. This ruins the performance of the aircraft since the nonlinearities are not exploited. The reinforcement learning control can take advantages of the coupling effect between the trajectory and the attitude dynamics and achieve optimized integrated

另一方面，飞行器飞行的控制通常将系统分为外环轨迹控制和内环姿态控制[18]。这破坏了飞行器的性能，因为没有利用非线性特性。强化学习控制可以利用轨迹和姿态动力学之间的耦合效应，实现优化的集成

flight control. In Ref. [19], the intelligent control for two different aerobatic maneuvers of the fixed-wing aircraft are developed by one variant of DQN, and it well explores and exploits the nonlinearities of dynamics. In this paper, the DDPG algorithm is employed to develop the 6 Degree-of-freedom (DOF) cruise controller for a full-sized fixed-wing aircraft. By directly learning the control law from the flight states to the aero-surfaces and thrust control, the intelligent integrated flight controller is developed and it avoids the separation of guidance and control, which is common during the traditional controller design.

飞行控制。在文献[19]中，通过 DQN 的一种变体，开发了固定翼飞机两种不同特技飞行的智能控制，它很好地探索和利用了动力学的非线性特性。本文采用 DDPG 算法为全尺寸固定翼飞机开发了 6 自由度（DOF）巡航控制器。通过直接从飞行状态学习到气动舵面和推力控制的控制律，开发了智能集成飞行控制器，避免了传统控制器设计中常见的制导与控制分离问题。

2 MATHEMATICAL MODEL

2 数学模型

The fixed-wing aircraft body coordinate frame b and the ground coordinate frame g are defined first [20]. The body frame b is fixed with the aircraft, with its origin o_b located at the center of mass. The $o_b x_b$ axis lies in the longitudinal symmetrical plane and points to the nose. The $o_b y_b$ axis is perpendicular to the symmetry plane and points to the right of the fuselage. The $o_b z_b$ axis lies in the symmetry plane and points down. The ground system g has the origin o_g that is located at some point on the ground. The $o_g x_g$ and $o_g y_g$ axes lie in the horizontal plane, pointing to the north and the east, respectively. The $o_g z_g$ axis is parallel to the vertical direction, towards the center of the earth.

首先定义固定翼飞机的机体坐标系 b 和地面坐标系 g [20]。机体坐标系 b 固定在飞机上，其原点 o_b 位于质心。 $o_b x_b$ 轴位于纵向对称平面内并指向机头。 $o_b y_b$ 轴垂直于对称平面并指向机身右侧。 $o_b z_b$ 轴位于对称平面内并指向下方。地面坐标系 g 的原点 o_g 位于地面某一点。 $o_g x_g$ 和 $o_g y_g$ 轴位于水平面内，分别指向北方和东方。 $o_g z_g$ 轴平行于垂直方向，指向地心。

Assume the thrust is along the $o_b x_b$ axis, the kinematic and dynamic equations for the aircraft trajectory motion are

假设推力沿 $o_b x_b$ 轴，飞机轨迹运动的运动学和动力学方程为

$$\dot{\boldsymbol{r}}=\begin{bmatrix}\dot{x}\\ \dot{y}\\ \dot{z}\end{bmatrix}=\boldsymbol{R}_b^g\begin{bmatrix}u\\ v\\ w\end{bmatrix}$$

$$\dot{\boldsymbol{v}}=\begin{bmatrix}\dot{u}\\ \dot{v}\\ \dot{w}\end{bmatrix}=\begin{bmatrix}vr-wq-gR_g^b(1,3)+\frac{X+\eta T_{\max}}{m}\\ -ur+wp-gR_g^b(2,3)+\frac{Y}{m}\\ uq-vp-gR_g^b(3,3)+\frac{Z}{m}\end{bmatrix}$$

where $\boldsymbol{r}=[x\quad y\quad z]^{\mathrm{T}}$ is the aircraft position vector in the ground reference frame, with " T " denoting the transpose operator, and $\boldsymbol{v}=[u\quad v\quad w]^{\mathrm{T}}$ is the aircraft velocity vector, described in the body frame. m is the mass. X, Y, and Z are the aerodynamic force, along the $o_b x_b, o_b y_b$, and $o_b z_b$ axis, respectively. η is the throttle of engine. g is the acceleration of gravity. $\boldsymbol{R}_b^g$ is the rotation matrix from the body frame b to the ground frame g.

其中 $\boldsymbol{r}=[x\quad y\quad z]^{\mathrm{T}}$ 是飞机在地面参考系中的位置矢量，“T”表示转置运算符，$\boldsymbol{v}=[u\quad v\quad w]^{\mathrm{T}}$ 是在机体坐标系中描述的飞机速度矢量。 m 是质量。 X, Y 和 Z 分别是沿 $o_b x_b, o_b y_b$ 和 $o_b z_b$ 轴的气动力。 η 是发动机的油门。 g 是重力加速度。 $\boldsymbol{R}_b^g$ 是从机体坐标系 b 到地面坐标系 g 的旋转矩阵。

The aircraft attitude kinematic and dynamic equations are

飞机姿态运动学和动力学方程为

$$\dot{\boldsymbol{q}}=\frac{1}{2}\begin{bmatrix}-q_1 & -q_2 & -q_3\\ q_0 & -q_3 & q_2\\ q_3 & q_0 & -q_1\\ -q_2 & q_1 & q_0\end{bmatrix}\boldsymbol{\omega}$$

$$\dot{\boldsymbol{\omega}}=\begin{bmatrix}\dot{p}\\ \dot{q}\\ \dot{r}\end{bmatrix}=\boldsymbol{I}^{-1}\left\{-\begin{bmatrix}0 & -r & q\\ r & 0 & -p\\ -q & p & 0\end{bmatrix}\boldsymbol{I}\begin{bmatrix}p\\ q\\ r\end{bmatrix}+\begin{bmatrix}l_a\\ m_a\\ n_a\end{bmatrix}\right\}$$

where $\boldsymbol{q}=[q_0\quad q_1\quad q_2\quad q_3]^{\mathrm{T}}$ is the attitude quaternion and $\boldsymbol{\omega}=[p\quad q\quad r]^{\mathrm{T}}$ is the angular velocity. $\boldsymbol{I}$ is the inertia matrix of the aircraft. l_a, m_a and n_a are the aerodynamic moments, along the $o_b x_b, o_b y_b$ and $o_b z_b$ axis, respectively.

其中 $\boldsymbol{q}=[q_0\quad q_1\quad q_2\quad q_3]^{\mathrm{T}}$ 是姿态四元数，$\boldsymbol{\omega}=[p\quad q\quad r]^{\mathrm{T}}$ 是角速度。 $\boldsymbol{I}$ 是飞行器的惯性矩阵。 l_a, m_a 和 n_a 分别是沿 $o_b x_b, o_b y_b$ 轴和 $o_b z_b$ 轴的气动力矩。

The aircraft considered in the following is a full-sized F/A-18 aircraft. The mass is $m=15119\ \mathrm{kg}$. The inertia components are $I_{xx}=31184\ \mathrm{kg\ m}^2, I_{yy}=205125\ \mathrm{kg\ m}^2$, $I_{zz}=230414\ \mathrm{kg\ m}^2$, and $I_{xz}=-4028.1\ \mathrm{kg}\cdot\ \mathrm{m}^2$. The aerodynamics are given in Ref. [21].

下文考虑的飞行器是一架全尺寸 F/A-18 飞行器。质量为 $m=15119\ \mathrm{kg}$ 。惯性分量为 $I_{xx}=31184\ \mathrm{kg\ m}^2, I_{yy}=205125\ \mathrm{kg\ m}^2$ 、 $I_{zz}=230414\ \mathrm{kg\ m}^2$ 和 $I_{xz}=-4028.1\ \mathrm{kg}\cdot\ \mathrm{m}^2$ 。气动数据见参考文献 [21]。

3 MODELING AND VERIFICATION OF NN CONTROLLER

3 NN 控制器的建模与验证

In order to investigate the generalization performance of the NN controller model, first the PID flight control law is designed and used to develop the NN controller through the supervised learning.

为了研究 NN 控制器模型的泛化性能，首先设计了 PID 飞行控制律，并通过监督学习开发了 NN 控制器。

3.1 PID Controller Design

3.1 PID 控制器设计

To realize the cruise flight of the aircraft, the longitudinal and lateral control laws are designed. The longitudinal control includes the speed control and the height control. The speed control law is

为实现飞行器的巡航飞行，设计了纵向和横向控制律。纵向控制包括速度控制和高度控制。速度控制律为

$$\eta=-k_{pV}\left(V-V_{\mathrm{cmd}}\right)-k_{iV}\int\left(V-V_{\mathrm{cmd}}\right)\mathrm{d}t$$

where V_{cmd} is the commanded aircraft speed and $V=\sqrt{u^2+v^2+w^2}$ is the aircraft speed. k_{pV} and k_{iV} are the corresponding proportional and integral control gains. The height control law is

其中 V_{cmd} 是指令飞行器速度， $V=\sqrt{u^2+v^2+w^2}$ 是飞行器速度。 k_{pV} 和 k_{iV} 是相应的比例和积分控制增益。高度控制律为

$$\delta_e=-k_q q-k_\theta\left(\theta-\theta_{\mathrm{cmd}}\right)$$

where δ_e is the elevator. k_q and k_θ are the proportional control gains for the pitch rate q and the pitch angle θ. θ_{cmd} is the pitch angle command and is given by

其中 δ_e 是升降舵。 k_q 和 k_θ 是俯仰角速率 q 和俯仰角 θ 的比例控制增益。 θ_{cmd} 是俯仰角指令，由下式给出

$$\theta_{\mathrm{cmd}}=-k_h\left(h-h_{\mathrm{cmd}}\right)-k_{V_z} V_z-k_{ih}\int\left(h-h_{\mathrm{cmd}}\right)\mathrm{d}t$$

where $h=-z$ is the height and h_{cmd} is the commanded height. V_z is the aircraft longitudinal vertical speed. k_h and k_{V_z} are the proportional gains regarding the height and the speed. k_{ih} is the integral gain of the height control.

其中 $h=-z$ 是高度， h_{cmd} 是指令高度。 V_z 是飞机的纵向垂直速度。 k_h 和 k_{V_z} 是关于高度和速度的比例增益。 k_{ih} 是高度控制的积分增益。

The lateral-directional control laws are

横航向控制律为

$$\delta_a=k_p p+k_\phi\left(\phi-\phi_{\mathrm{cmd}}\right)+k_{i\phi}\int\left(\phi-\phi_{\mathrm{cmd}}\right)\mathrm{d}t$$
$$\delta_r=k_r r-k_\beta \beta$$

where ϕ is the aircraft roll angle and β is the aircraft sideslip angle. k_p, k_ϕ, k_r, and k_β are the proportional control gains for the corresponding variables. $k_{i\phi}$ is the integral control gain. The roll angle command ϕ_{cmd} is

其中 ϕ 是飞机滚转角， β 是飞机侧滑角。 k_p, k_ϕ, k_r 和 k_β 是相应变量的比例控制增益。 $k_{i\phi}$ 是积分控制增益。滚转角指令 ϕ_{cmd} 为

$$\phi_{\mathrm{cmd}}=-k_\psi\left(\psi-\psi_{\mathrm{cmd}}\right)$$

where ψ is the yaw angle, and ψ_{cmd} is the commanded yaw angle, which represents the desired heading when there is no sideslip. k_ψ is the proportional control gain.

其中 ψ 是偏航角， ψ_{cmd} 是指令偏航角，表示无侧滑时的期望航向。 k_ψ 是比例控制增益。

3.2 NN Controller Training Through Supervised Learning

3.2 通过监督学习训练神经网络控制器

The observations for the aircraft flight control are given by a one-dimensional vector, and the NN model with fully connected feedforward multi-hidden layers is suitable for the modeling of the controller. In order to exam the generalization ability of the NN model, the supervised learning is implemented. With Pytorch, the developed NN controller model has 5 layers. The input layer has 10 input units, including the height, the velocity vector, the Euler angles (transformed from the quaternion) and the angular velocity vector. The output layer has 4 output units, corresponding to the aileron, elevator, rudder and throttle. There are 3 hidden layers in the NN model, and the number of units are set as 32, 64 and 128, respectively. The first 4 layers use “Relu” activation function while the output layer uses “Tanh” activation function.

飞行器飞控的观测值由一维向量给出，具有全连接前馈多隐藏层的神经网络模型适用于控制器的建模。为了检验神经网络模型的泛化能力，我们实现了监督学习。使用 PyTorch，开发的神经网络控制器模型有 5 层。输入层有 10 个输入单元，包括高度、速度向量、欧拉角（由四元数转换而来）和角速度向量。输出层有 4 个输出单元，分别对应副翼、升降舵、方向舵和油门。神经网络模型中有 3 个隐藏层，单元数分别设置为 32、64 和 128。前 4 层使用“Relu”激活函数，而输出层使用“Tanh”激活函数。

Take the PID controller as “teacher”; the NN controller is trained through the labeled data generated in flight simulations with different initial conditions. The network parameters are initialized by the “xavier_uniform” function, and the “Adam” method is used for the optimization [22]. In the supervised learning, the learning rate is 1×10^{-3}, the batch size is 128, and the number of training iteration is 20000. The trained NN controller is applied to the flight simulation. Starting from the origin of the ground frame, the aircraft has an initial speed of $100 \mathrm{~m/s}$ and the initial heading angle is 0°. The commanded height is 100 m . The commanded speed is $76.32 \mathrm{~m/s}$ and the commanded heading angle is 31.61°. Simulation results under the PID controller and the NN controller are presented in Figs. 1-3. Figure 1 plots the height curve of the aircraft and Fig. 2 gives the velocity profiles. In Fig. 3, the control commands are compared. From these figures, it is shown that the performance of the NN controller is very similar to that of the PID controller. This proves that the NN model meets the modeling requirement for the intelligent controller design.

以 PID 控制器为“教师”，通过在不同初始条件下的飞行仿真中生成的带标签数据来训练神经网络控制器。网络参数通过“xavier_uniform”函数初始化，“Adam”方法用于优化 [22]。在监督学习中，学习率为 1×10^{-3} ，批大小为 128，训练迭代次数为 20000。训练好的神经网络控制器应用于飞行仿真。从地面坐标系原点开始，飞行器初始速度为 $100 \mathrm{~m/s}$ ，初始航向角为 0° 。指令高度为 100 米。指令速度为 $76.32 \mathrm{~m/s}$ ，指令航向角为 31.61° 。PID 控制器和神经网络控制器下的仿真结果如图 1-3 所示。图 1 绘制了飞行器的高度曲线，图 2 给出了速度剖面。图 3 比较了控制指令。从这些图中可以看出，神经网络控制器的性能与 PID 控制器的性能非常相似。这证明了神经网络模型满足智能控制器设计的建模要求。

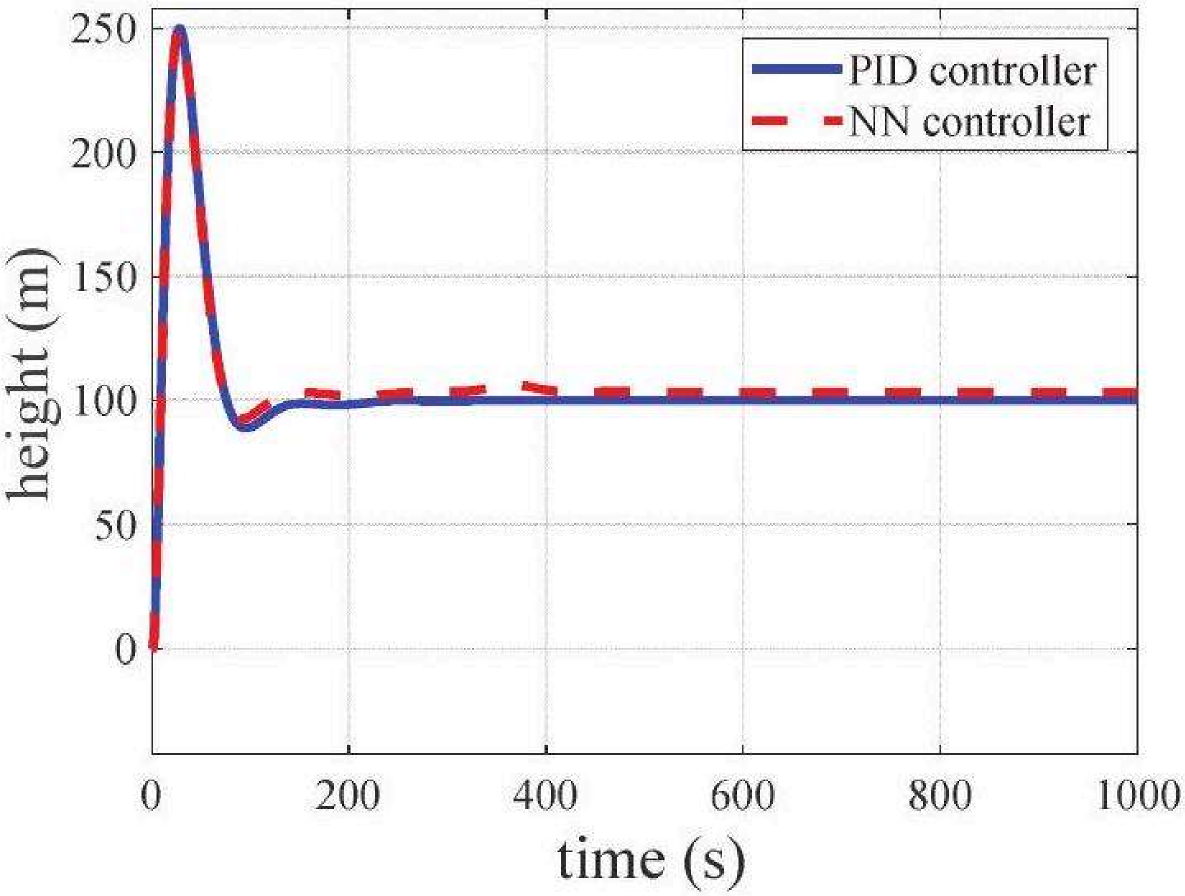

Figure 1: Fig. 1 Height curves under the PID controller and the NN controller.

图 1：图 1 PID 控制器和神经网络控制器下的高度曲线。

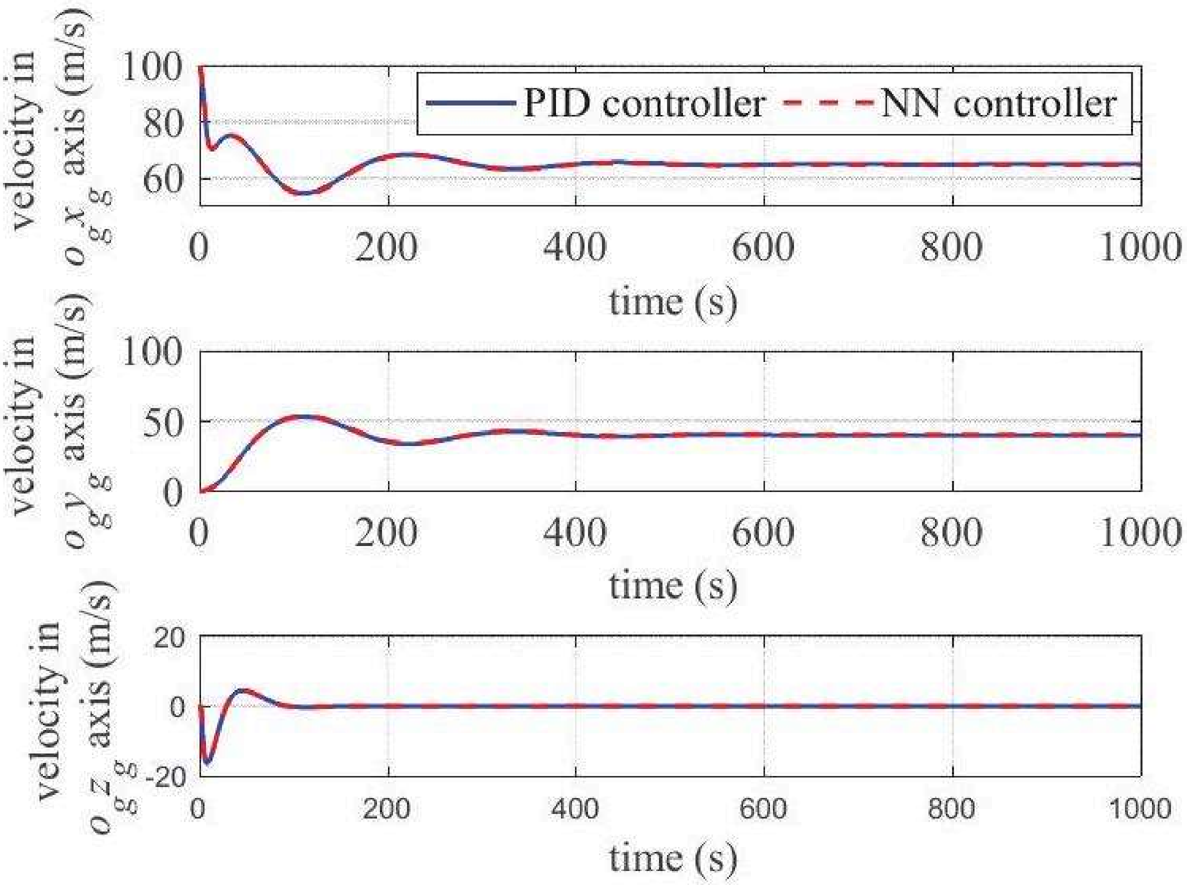

Figure 2: Fig. 2 Velocity profiles under the PID controller and the NN controller.

图 2：图 2 PID 控制器和 NN 控制器下的速度曲线。

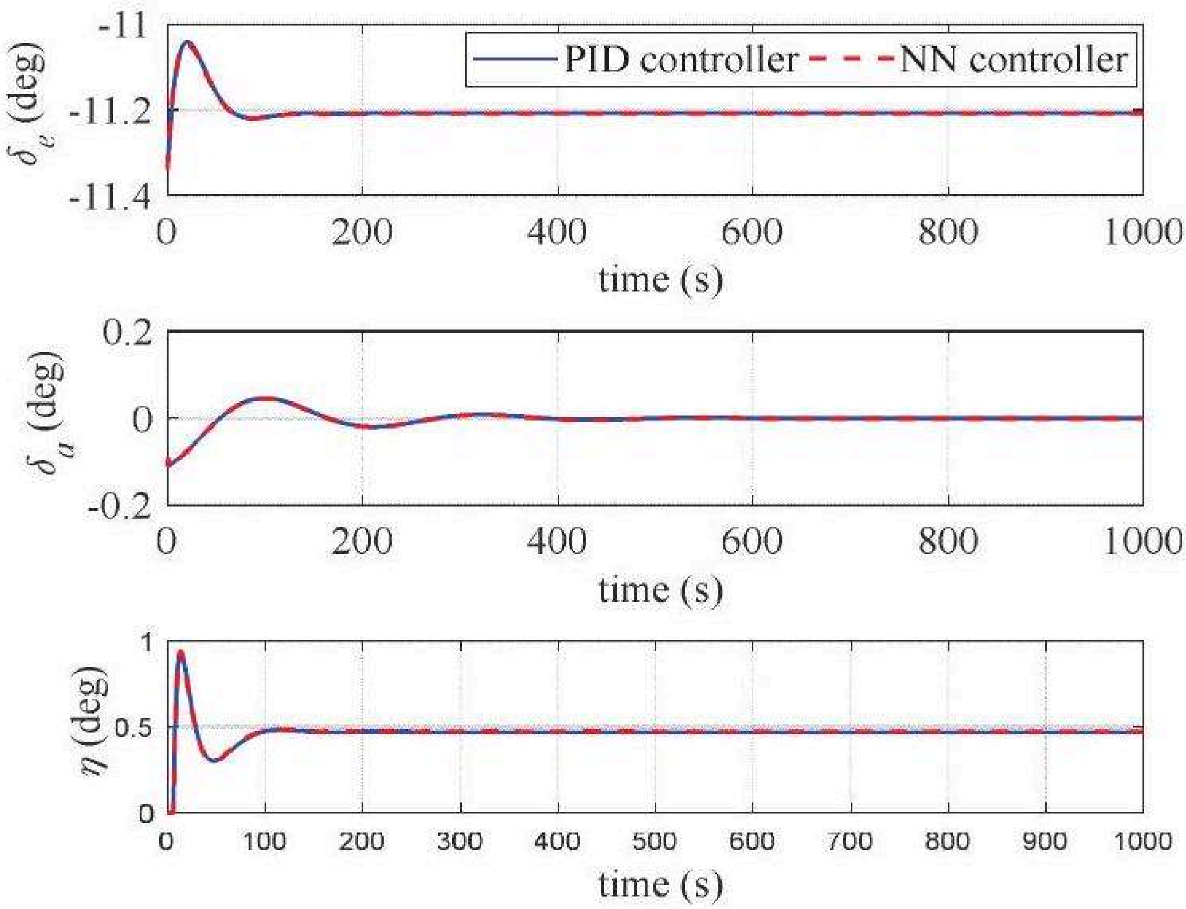

Figure 3: Fig. 3 Control profiles under the PID controller and the NN controller.

图 3：图 3 PID 控制器和 NN 控制器下的控制曲线。

4 NN CONTROLLER DESIGN BY REINFORCEMENT LEARNING

4 基于强化学习的神经网络控制器设计

With the improvement of algorithms and the development of hardware capability, the training of deep NN models is no longer the bottleneck that restricts the studies on deep (reinforcement) learning [23]. Thus, in the following the NN controller is trained through directly interacting with the environment from a random initialization of NN parameters. By exploring the simulation data, the intelligent controller may exploit the coupling effect of the nonlinear dynamics and achieve end-to-end integrated optimal control.

随着算法的改进和硬件能力的提升，深度神经网络模型的训练不再是限制深度（强化）学习研究的瓶颈[23]。因此，在下文中，神经网络控制器通过与环境直接交互，从神经网络参数的随机初始化开始进行训练。通过探索仿真数据，智能控制器可以利用非线性动力学的耦合效应，实现端到端集成优化控制。

4.1 DDPG Algorithm

4.1 DDPG 算法

The DDPG algorithm is developed to solve the “continuous action” type problems, upon the DQN algorithm. It uses the Actor-Critic architecture, where the Critic evaluates the action value function of the policy, i.e., $Q = Q(\boldsymbol{x}, \boldsymbol{u})$, and the Actor defines a deterministic policy function $\boldsymbol{u} = \pi(\boldsymbol{x})$. With the NN model to approximate these functions, the Critic NN is expressed by $\tilde{Q}\left(\boldsymbol{x}, \boldsymbol{u}; \boldsymbol{\theta}_Q\right)$, with $\boldsymbol{\theta}_Q$ denoting

DDPG 算法是在 DQN 算法的基础上，为解决“连续动作”类型问题而开发的。它采用 Actor-Critic 架构，其中 Critic 评估策略的动作值函数，即 $Q = Q(\boldsymbol{x}, \boldsymbol{u})$ ，而 Actor 定义确定性策略函数 $\boldsymbol{u} = \pi(\boldsymbol{x})$ 。使用神经网络模型来近似这些函数，Critic 神经网络表示为 $\tilde{Q}\left(\boldsymbol{x}, \boldsymbol{u}; \boldsymbol{\theta}_Q\right)$ ，其中 $\boldsymbol{\theta}_Q$ 表示

the parameters of the Critic NN; the Actor NN is expressed by $\tilde{\pi}\left(\boldsymbol{x}; \boldsymbol{\theta}_u\right)$, with $\boldsymbol{\theta}_u$ the NN parameters. Since the agent may learn the optimal policy regardless of the employed policy, the DDPG algorithm falls into the category of the off-policy method. The update law of the Critic NN parameters is similar to that of DQN, established by reducing the following performance index

Critic 神经网络的参数；Actor 神经网络表示为 $\tilde{\pi}\left(\boldsymbol{x}; \boldsymbol{\theta}_u\right)$ ，其中 $\boldsymbol{\theta}_u$ 是神经网络参数。由于智能体可以学习最优策略，而无论采用何种策略，DDPG 算法属于离策略方法。Critic 神经网络参数的更新法则类似于 DQN，通过减小以下性能指标建立

$$J = (\Delta Q)^2 = \left(y_t - \tilde{Q}\left(\boldsymbol{x}_t, \boldsymbol{u}_t; \boldsymbol{\theta}_Q\right)\right)^2$$

where y_t is the estimation of the action value, $\tilde{Q}\left(\boldsymbol{x}_t, \boldsymbol{u}_t; \boldsymbol{\theta}_Q\right)$ is the prediction of the action value, and ΔQ is their difference. Hence the update law for the Critic NN parameters is

其中 y_t 是动作值的估计， $\tilde{Q}\left(\boldsymbol{x}_t, \boldsymbol{u}_t; \boldsymbol{\theta}_Q\right)$ 是动作值的预测， ΔQ 是它们的差值。因此，Critic 神经网络参数的更新法则为

$$\left(\boldsymbol{\theta}_Q\right)_{k+1} = \left(\boldsymbol{\theta}_Q\right)_k + \alpha \Delta Q \frac{\partial Q}{\partial (\boldsymbol{\theta}_Q)_k}$$

where α is the learning rate of the Critic network. On the other hand, in order to make the outputs of the Actor network maximize the output of the Critic network, the gradient of the action value function with respect to the Actor network parameters may be derived as $\frac{\partial Q}{\partial u}\frac{\partial u}{\partial \theta_u}$, and then the update law for the Actor network parameters is

其中 α 是 Critic 网络的学习率。另一方面，为了使 Actor 网络的输出最大化 Critic 网络的输出，动作值函数对 Actor 网络参数的梯度可以推导为 $\frac{\partial Q}{\partial u}\frac{\partial u}{\partial \theta_u}$ ，然后 Actor 网络参数的更新规则为

$$\left(\boldsymbol{\theta}_u\right)_{k+1} = \left(\boldsymbol{\theta}_u\right)_k + \beta \frac{\partial Q}{\partial u} \frac{\partial u}{\partial (\boldsymbol{\theta}_u)_k}$$

where β is the learning rate of the Actor network.

其中 β 是 Actor 网络的学习率。

Like the DQN, the DDPG algorithm also considers the target networks that make the training stable, including the Target-Critic network $\tilde{Q}\left(\boldsymbol{x}, \boldsymbol{u}; \boldsymbol{\theta}'_Q\right)$ and the Target-Actor network $\tilde{\boldsymbol{\pi}}\left(\boldsymbol{x}; \boldsymbol{\theta}'_u\right)$. They have the same structure to the original networks, and the corresponding parameters are denoted by $\boldsymbol{\theta}'_Q$ and $\boldsymbol{\theta}'_u$, respectively. With the target networks, reinforcement learning behaves similarly to the supervised learning, and more stable learning process may be achieved. This is beneficial to improve the convergence of NN parameters. The target network parameters adopt a gradual manner update law, that is

与 DQN 类似，DDPG 算法也考虑了使训练稳定的目标网络，包括目标 Critic 网络 $\tilde{Q}\left(\boldsymbol{x},\boldsymbol{u};\boldsymbol{\theta}'_Q\right)$ 和目标 Actor 网络 $\tilde{\boldsymbol{\pi}}\left(\boldsymbol{x};\boldsymbol{\theta}'_u\right)$。它们与原始网络具有相同的结构，相应的参数分别表示为 $\boldsymbol{\theta}'_Q$ 和 $\boldsymbol{\theta}'_u$。有了目标网络，强化学习的行为类似于监督学习，可以实现更稳定的学习过程。这有利于提高神经网络参数的收敛性。目标网络参数采用渐进式更新规则，即

$$\boldsymbol{\theta}'_Q=\tau\boldsymbol{\theta}_Q+(1-\tau)\boldsymbol{\theta}'_Q$$
$$\boldsymbol{\theta}'_u=\tau\boldsymbol{\theta}_u+(1-\tau)\boldsymbol{\theta}'_u$$

where τ is the update rate. Obviously the larger τ is, the faster the update will be. In addition, the DDPG algorithm also introduces the experience pool, which is advantageous to attenuate the correlation of data and address the non-stationary distribution of samples through mini-batch sampling. Consider the target networks and the mini-batch sampling, Eq. (11) needs to be adapted as

其中 τ 是更新率。显然， τ 越大，更新越快。此外，DDPG 算法还引入了经验池，通过小批量采样，有利于减弱数据的相关性，解决样本的非平稳分布问题。考虑到目标网络和小批量采样，公式 (11) 需要修改为

$$J=\sum_N(\Delta Q)^2=\sum_N\left(y_t-\tilde{Q}\left(\boldsymbol{x}_t,\boldsymbol{u}_t;\boldsymbol{\theta}_Q\right)\right)^2$$

where N is the size of the mini-batch. The action value function is estimated as

其中 N 是 mini-batch 的大小。动作值函数估计为

$$y_t=r_t+\gamma\tilde{Q}\left(\boldsymbol{x}_{t+1},\boldsymbol{u}_{t+1};\boldsymbol{\theta}'_Q\right)$$

where r_t is the reward function and γ is the discount factor. Correspondingly, the gradients on the parameters in Eqs.

其中 r_t 是奖励函数， γ 是折扣因子。相应地，公式中的参数梯度

(12) and (13) need to be modified as the sum on the whole mini-batch data set.

(12) 和 (13) 需要修改为整个 mini-batch 数据集上的总和。

In particular, in order to balance the exploration and exploitation mechanism, the control output by the Actor network will be blended with random noise during the learning.

特别是，为了平衡探索和利用机制，Actor 网络输出的控制将在学习过程中与随机噪声混合。

4.2 NN Controller Learning

4.2 神经网络控制器学习

For the cruise control of the fixed-wing aircraft, in the reinforcement learning the commanded height is h_c, while the commanded velocity is given in the ground frame, namely, $[v_{xc}\quad v_{yc}\quad 0]^{\mathrm{T}}=[V_{\mathrm{cmd}}\cos\psi_{\mathrm{cmd}}\quad V_{\mathrm{cmd}}\sin\psi_{\mathrm{cmd}}\quad 0]^{\mathrm{T}}$. Analogous to the integral term in the PID controller that eliminates the steady-state error, the Actor network has 13 inputs; they are the height error, the three-dimensional velocity error, the Euler angles, the angular velocity, the integral of height error, and the integrals of velocity error along the $o_g x_g$ and $o_g y_g$ axes. The 4 action outputs are aileron, elevator, rudder and throttle, respectively. For the Critic network, there are 17 inputs, including the 13 state inputs of the Actor network and the 4 action inputs. The only 1 output is the action value. According to the expected control objective, the reward function at time t_k is set as

对于固定翼飞机的巡航控制，在强化学习中，指令高度为 h_c ，而指令速度在地面坐标系中给出，即 $[v_{xc} \quad v_{yc} \quad 0]^{\mathrm{T}} = [V_{\mathrm{cmd}}\cos\psi_{\mathrm{cmd}} \quad V_{\mathrm{cmd}}\sin\psi_{\mathrm{cmd}} \quad 0]^{\mathrm{T}}$ 。类似于 PID 控制器中消除稳态误差的积分项，Actor 网络有 13 个输入；它们是高度误差、三维速度误差、欧拉角、角速度、高度误差的积分，以及沿 $o_g x_g$ 和 $o_g y_g$ 轴的速度误差积分。4 个动作输出分别是副翼、升降舵、方向舵和油门。对于 Critic 网络，有 17 个输入，包括 Actor 网络的 13 个状态输入和 4 个动作输入。唯一的 1 个输出是动作值。根据预期的控制目标，时间 t_k 的奖励函数设置为

$$
\begin{aligned}
r_k(\boldsymbol{x},\boldsymbol{u}) = & w_h\left|h_k-h_c\right| + w_{vx}\left|(v_x)_k-v_{xc}\right| + w_{wy}\left|(v_y)_k-v_{yc}\right| \\
& + w_{vz}\left|(v_z)_k\right| + w_\beta|\beta| + w_\phi\left|\phi_k\right| + w_\psi\left|\psi_k-\arctan\left(v_{yc}/v_{xc}\right)\right| \\
& + w_p\left|p_k\right| + w_q\left|q_k\right| + w_r\left|r_k\right| \\
& + w_{ih}\left|\int(h_k-h_c)\mathrm{d}t\right| + w_{ixx}\left|\int((v_x)_k-v_{xc})\mathrm{d}t\right| + w_{iyy}\left|\int\left((v_y)_k-v_{yc}\right)\mathrm{d}t\right|
\end{aligned}
$$

where $w_i\,(i=h,v_x,v_y,v_z,\beta,\phi,\psi,p,q,r,ih,iv_x,iv_y)$ are weight coefficients. In learning the control law, an episode is defined as a finite horizon including a certain number of control periods of ΔT. The discount factor is $\gamma=0.99$, and then the return function is $R=\sum\gamma^k r_k$.

其中 $w_i\,(i=h,v_x,v_y,v_z,\beta,\phi,\psi,p,q,r,ih,iv_x,iv_y)$ 是权重系数。在学习控制律时，一个回合被定义为一个有限的时间范围，包括一定数量的 ΔT 控制周期。折扣因子是 $\gamma=0.99$ ，然后回报函数是 $R=\sum\gamma^k r_k$ 。

Use Pytorch to build the NN model. The structure of the Actor network is the similar to that of Sec. 3.2, except the input layer. The Critic network has 4 layers. There are 2 hidden layers, with the number of units being 64 and 128, respectively. For the first 3 layers, the activation function is “Relu” function, while the activation function is linear for the output layer. Both networks are initialized with the “xavier_uniform” function. The mini-batch size is $N=128$ and the length of an episode is $5000\Delta T$, with $\Delta T=0.1$ s. In addition, if the aircraft states exceed some preset boundary value, the episode will terminate immediately with penalty on the rewards. The weight coefficients in the reward function (18) are $w_i=0.01(i=h), w_i=0.01\,(i=v_x,v_y,v_z)$, $w_i=2(i=\beta,\phi,\psi), w_i=1(i=p,q,r)$, and $w_i=0.05\,(i=ih,iv_x,iv_y)$. After an episode ends, the Critic network and Actor network parameters are updated with the “Adam” optimization method. The learning rate are taken as $\alpha=1\times10^{-3}$ and $\beta=5\times10^{-3}$, respectively, and then the soft update for the target networks are performed, with a update rate of $\tau=0.001$.

使用 Pytorch 构建神经网络模型。Actor 网络的结构与 3.2 节的结构相似，除了输入层。Critic 网络有 4 层。有 2 个隐藏层，单元数分别为 64 和 128。对于前 3 层，激活函数是“Relu”函数，而输出层的激活函数是线性的。两个网络都用“xavier_uniform”函数初始化。小批量大小是 $N=128$ ，一个回合的长度是 $5000\Delta T$ ，其中 $\Delta T=0.1$ s 。此外，如果飞机状态超过一些预设的边界值，该回合将立即终止，并对奖励进行惩罚。奖励函数 (18) 中的权重系数是 $w_i=0.01(i=h), w_i=0.01\,(i=v_x,v_y,v_z)$ 、 $w_i=2(i=\beta,\phi,\psi), w_i=1(i=p,q,r)$ 和 $w_i=0.05\,(i=ih,iv_x,iv_y)$ 。一个回合结束后，Critic 网络和 Actor 网络的参数用“Adam”优化方法更新。学习率分别取 $\alpha=1\times10^{-3}$ 和 $\beta=5\times10^{-3}$ ，然后对目标网络进行软更新，更新率为 $\tau=0.001$ 。

The Python platform “Spyder” is employed to carry out the training of the NN controller. The training results are presented in Figs. 4-6. In Fig. 4, the average return function against the number of episodes is plotted. The length of episodes is given in Fig. 5 and the smoothed terminal error during the training is given in Fig. 6. It may be found that the performance of the NN controller is not improved monotonically. However, as the number of the training increases, the results show an overall improvement on the number of times when the aircraft successfully finishes the episodes. Compared with the quadrotor, the reinforcement learning control of the fixed-wing aircraft is more complicated, due to the balance between the aerodynamic force and aerodynamic moment. Moreover, for the full-sized aircraft, the training time is also greatly increased compared with a small aircraft.

Python 平台“Spyder”用于执行 NN 控制器的训练。训练结果如图 4-6 所示。图 4 绘制了平均回报函数与回合数的关系。图 5 给出了回合的长度，图 6 给出了训练期间平滑的终端误差。可以发现 NN 控制器的性能并非单调提升。然而，随着训练次数的增加，结果显示飞机成功完成回合的次数总体有所改善。与四旋翼飞行器相比，固定翼飞机的强化学习控制由于需要平衡气动力和气动力矩而更为复杂。此外，对于全尺寸飞机，训练时间也比小型飞机大大增加。

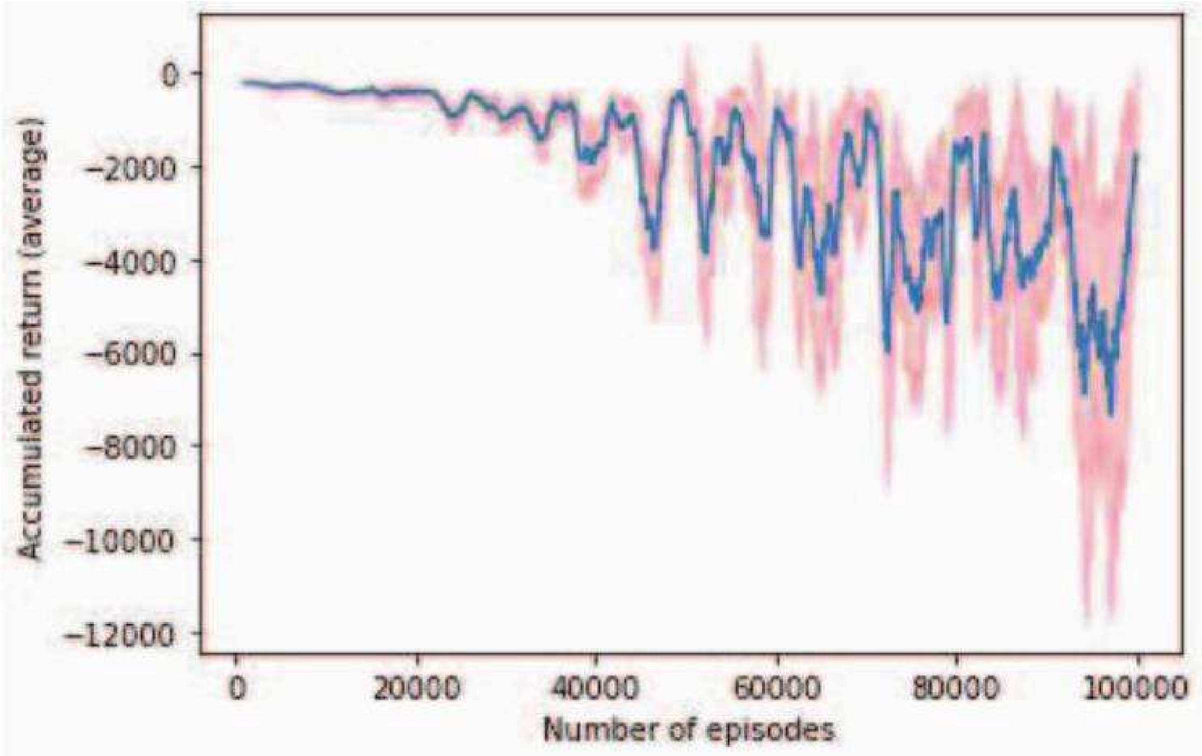

Figure 4: Fig. 4 Accumulated return against number of episodes in the reinforcement learning.

图 4：图 4 强化学习中累积回报与回合数的关系。

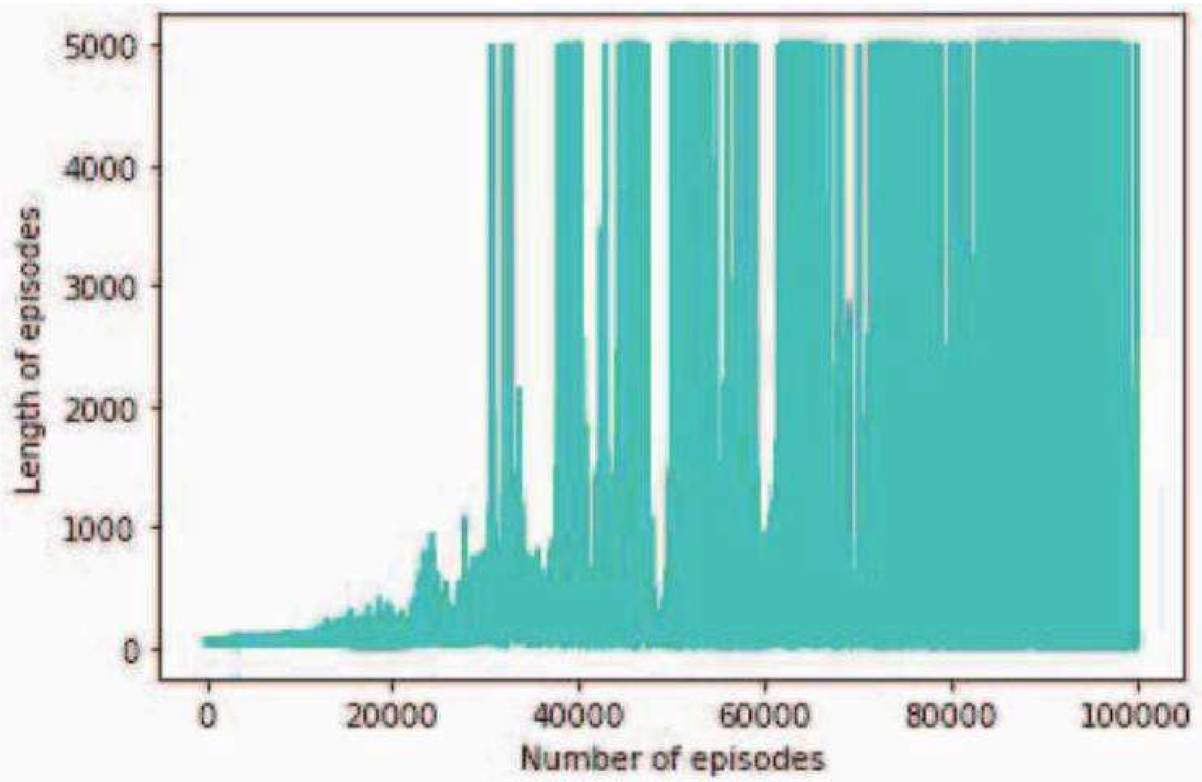

Figure 5: Fig. 5 Length of episodes against number of episodes in the reinforcement learning.

图 5：图 5 强化学习中回合长度与回合数的关系。

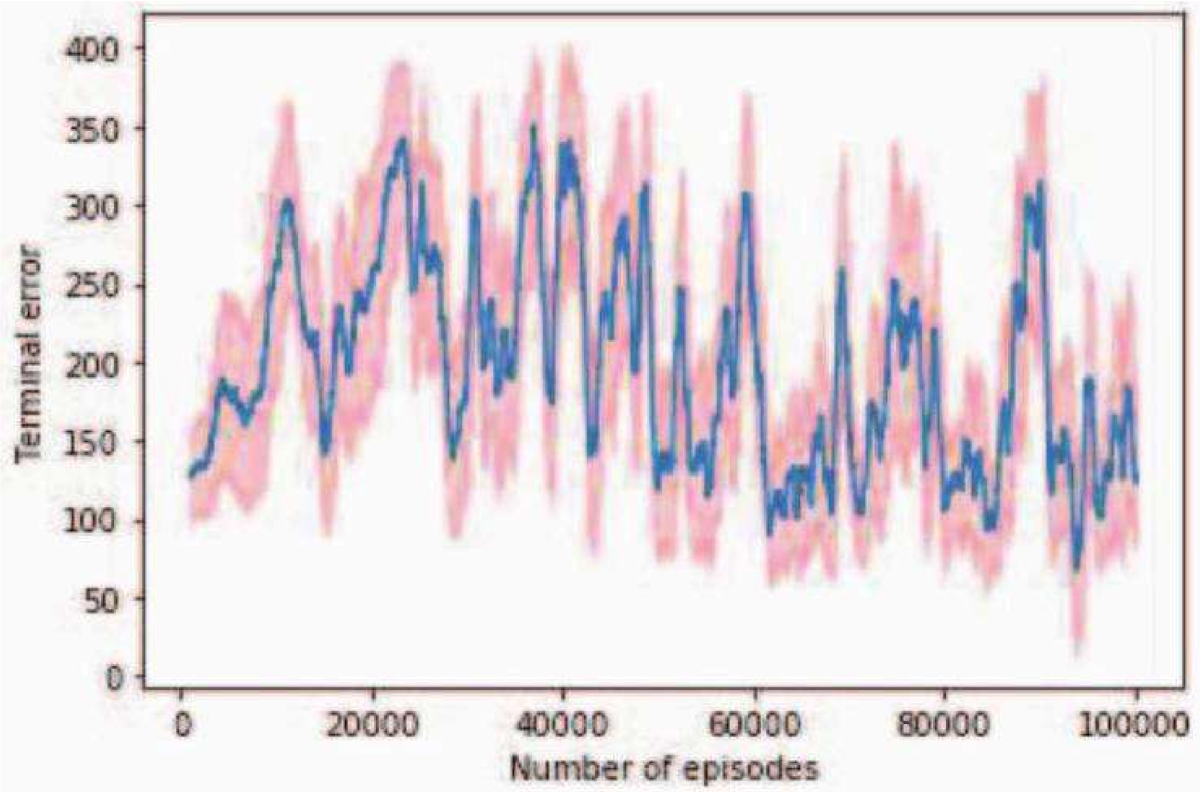

Figure 6: Fig. 6 Terminal error against number of episodes in the reinforcement learning.

图 6：图 6 强化学习中终端误差与回合数的关系。

5 FLIGHT SIMULATION

5 飞行模拟

Since the monotonous improvement on the performance of the NN controller cannot be guaranteed in the reinforcement learning, it may occur that the control effect of the trained NN controller deteriorates or the controller even destabilizes the aircraft as the training number increases. To get the NN controller that has desirable performance, the terminal errors relative to the target states were compared after each training episode and the NN controller with smaller error was selected.

由于强化学习无法保证神经网络控制器性能的单调提升，因此可能会出现随着训练次数的增加，训练好的神经网络控制器的控制效果变差，甚至导致飞行器失稳的情况。为了获得性能理想的神经网络控制器，在每次训练结束后，都会比较相对于目标状态的最终误差，并选择误差较小的神经网络控制器。

Upon the NN controller obtained after 78890 episodes of training, the flight simulation, with a time horizon of 1000s, was carried out on “Spyder”. The initial conditions and the target states are the same as those in Sec. 3. Figure 7 shows the horizontal trajectory of the aircraft, and the aircraft realizes the commanded heading. Figure 8 gives the height of the aircraft. It is shown that the height of the aircraft converges to the commanded value of 100 m after 50 s and the terminal error is 0 m . The velocity profiles of the aircraft are given in Fig. 9. After a transition period of 40 s , the aircraft approaches the commanded velocity and the commanded values are achieved exactly. Figure 10 plots the attitude Euler angles during the flight. It is shown that there is a steady roll angle of -5.84° during the cruise flight. The angular velocities of the aircraft are plotted in Fig. 12, and they all tend to zero after 40s.

在经过 78890 次训练后获得的神经网络控制器上，在“Spyder”上进行了时间范围为 1000 秒的飞行仿真。初始条件和目标状态与第 3 节中的相同。图 7 显示了飞机的水平轨迹，飞机实现了指令航向。图 8 给出了飞机的高度。结果表明，飞机高度在 50 秒后收敛到指令值 100 米，最终误差为 0 米。飞机的速度剖面如图 9 所示。经过 40 秒的过渡期后，飞机接近指令速度，并精确达到指令值。图 10 绘制了飞行过程中的姿态欧拉角。结果表明，在巡航飞行中存在一个稳定的滚转角 -5.84° 。飞机的角速度如图 12 所示，它们都在 40 秒后趋于零。

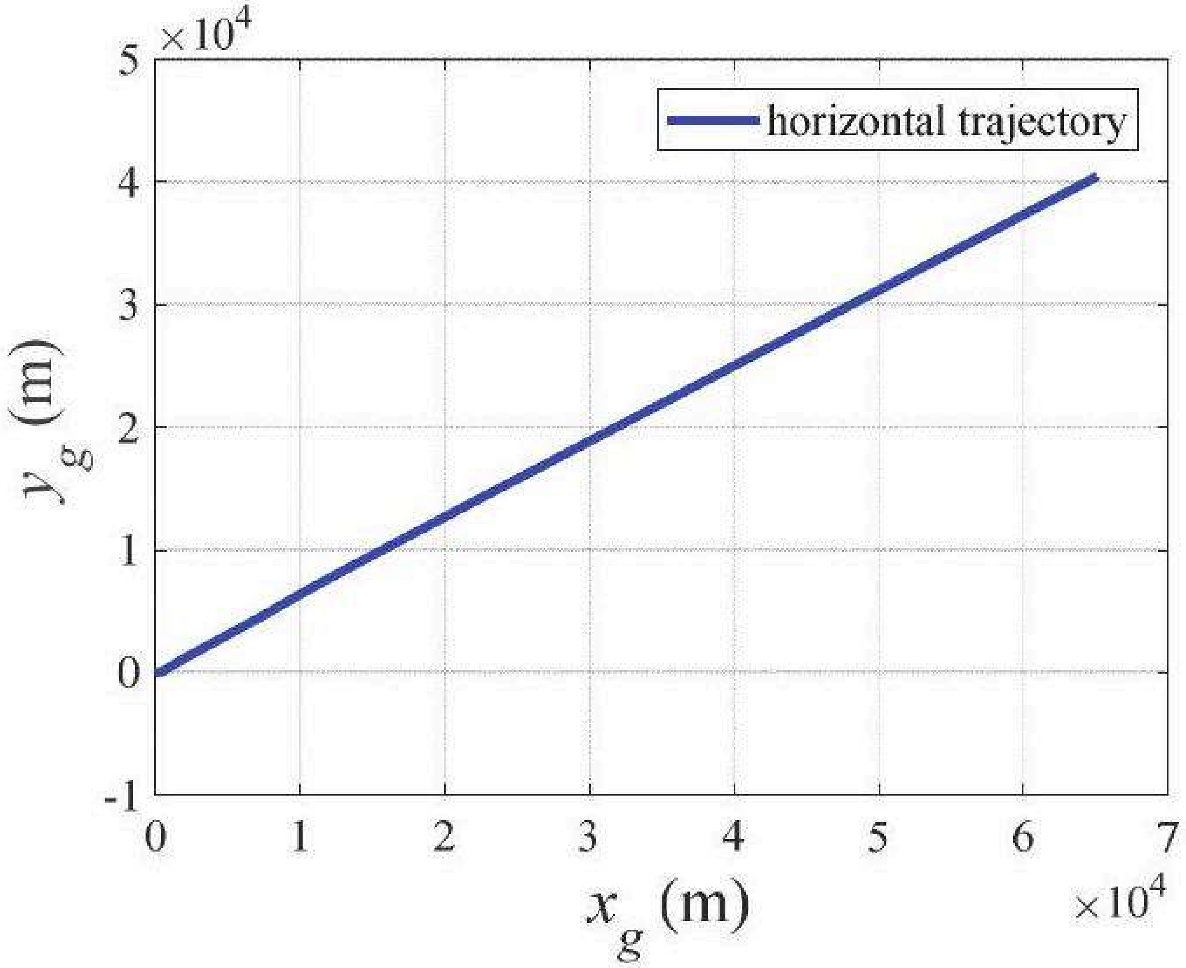

Figure 7: Fig. 7 Aircraft horizontal trajectory under the reinforcement learning NN controller.

图 7：图 7 强化学习神经网络控制器下的飞机水平轨迹。

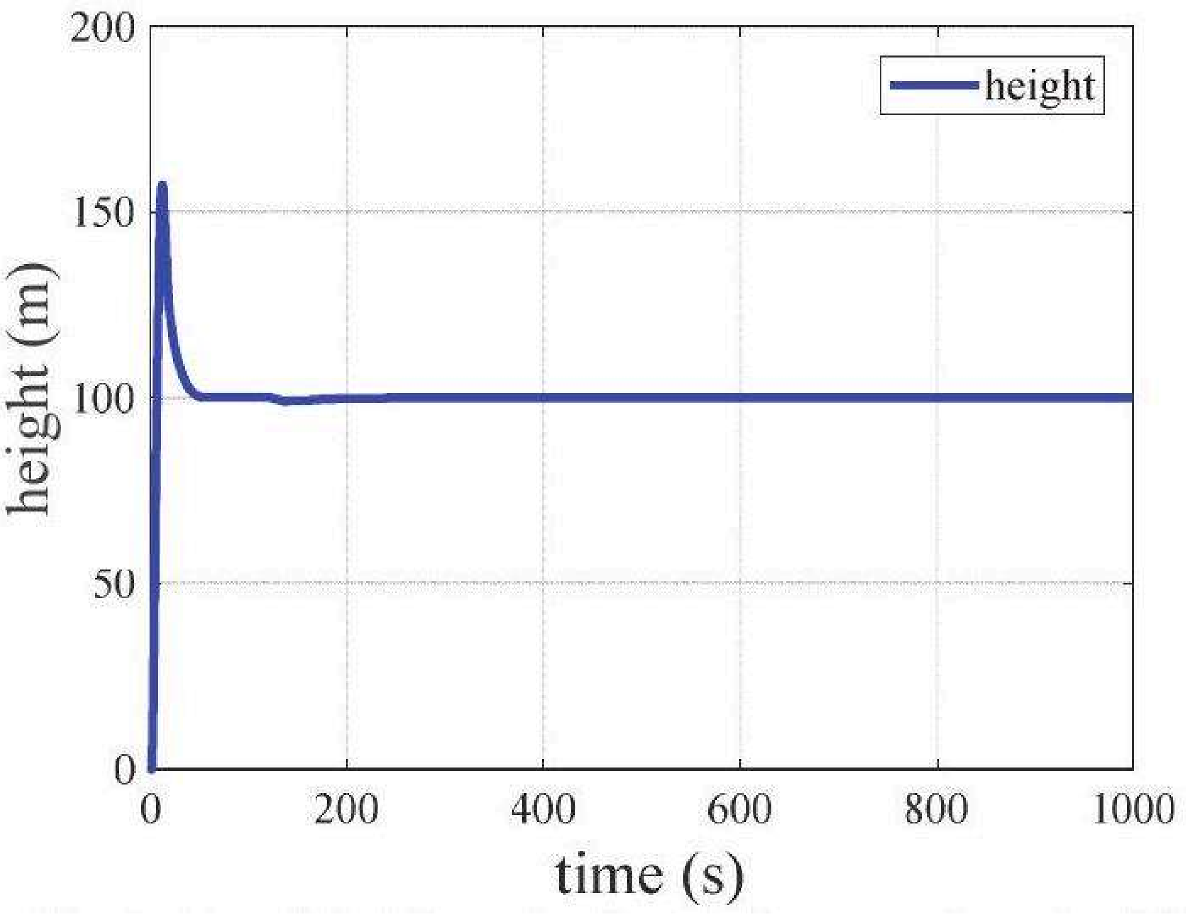

Figure 8: Fig. 8 Aircraft height under the reinforcement learning NN controller.

图 8：强化学习神经网络控制器下的飞机高度。

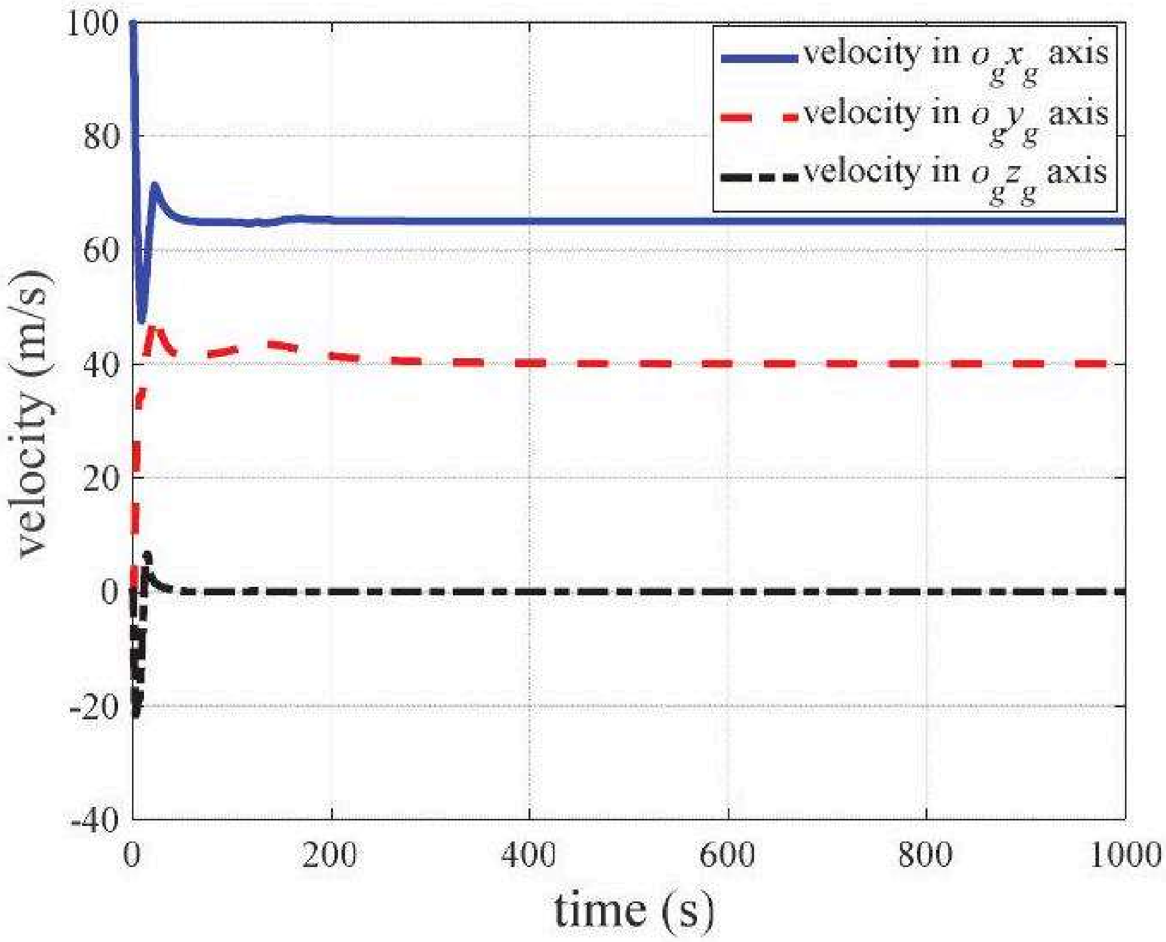

Figure 9: Fig. 9 Aircraft velocity under the reinforcement learning NN controller.

图 9：图 9 强化学习神经网络控制器下的飞机速度。

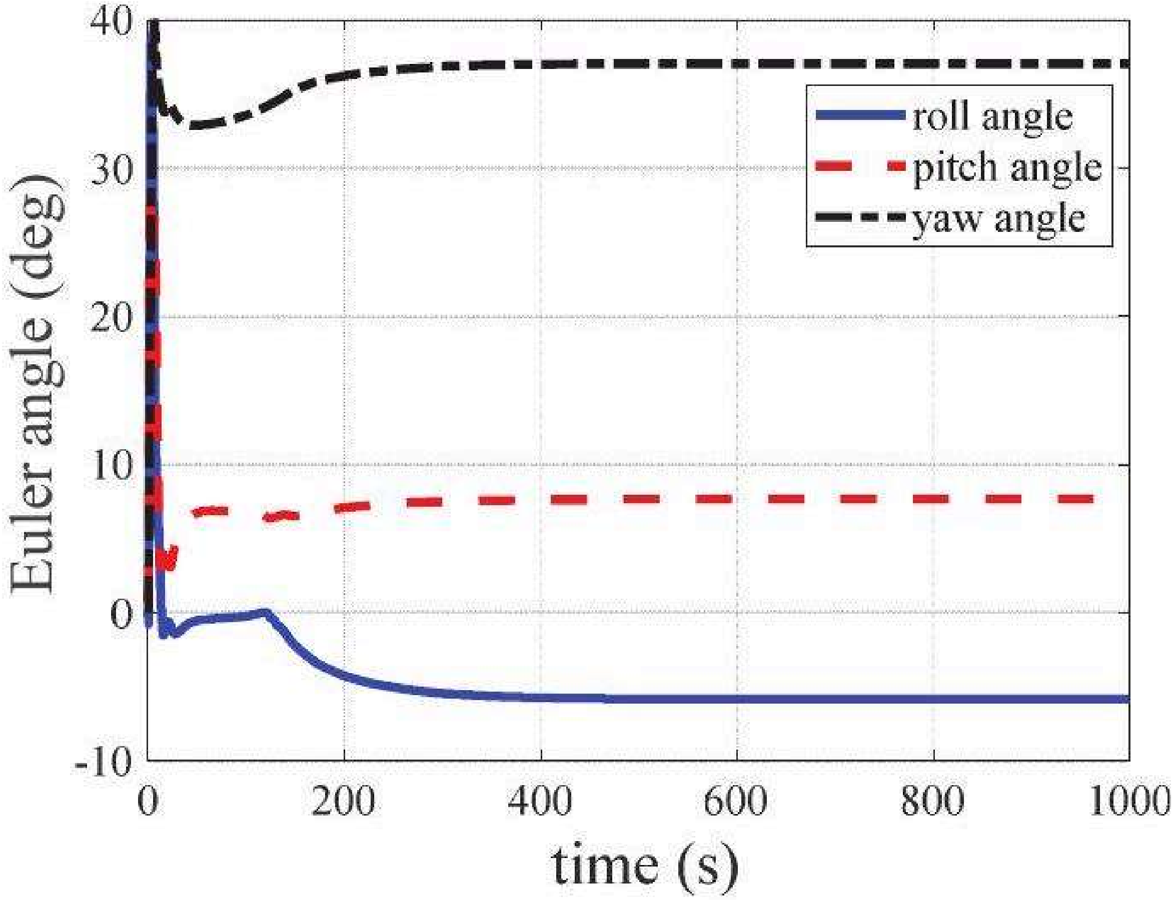

Figure 10: Fig. 10 Aircraft attitude under the reinforcement learning NN controller.

图 10：图 10 强化学习神经网络控制器下的飞机姿态。

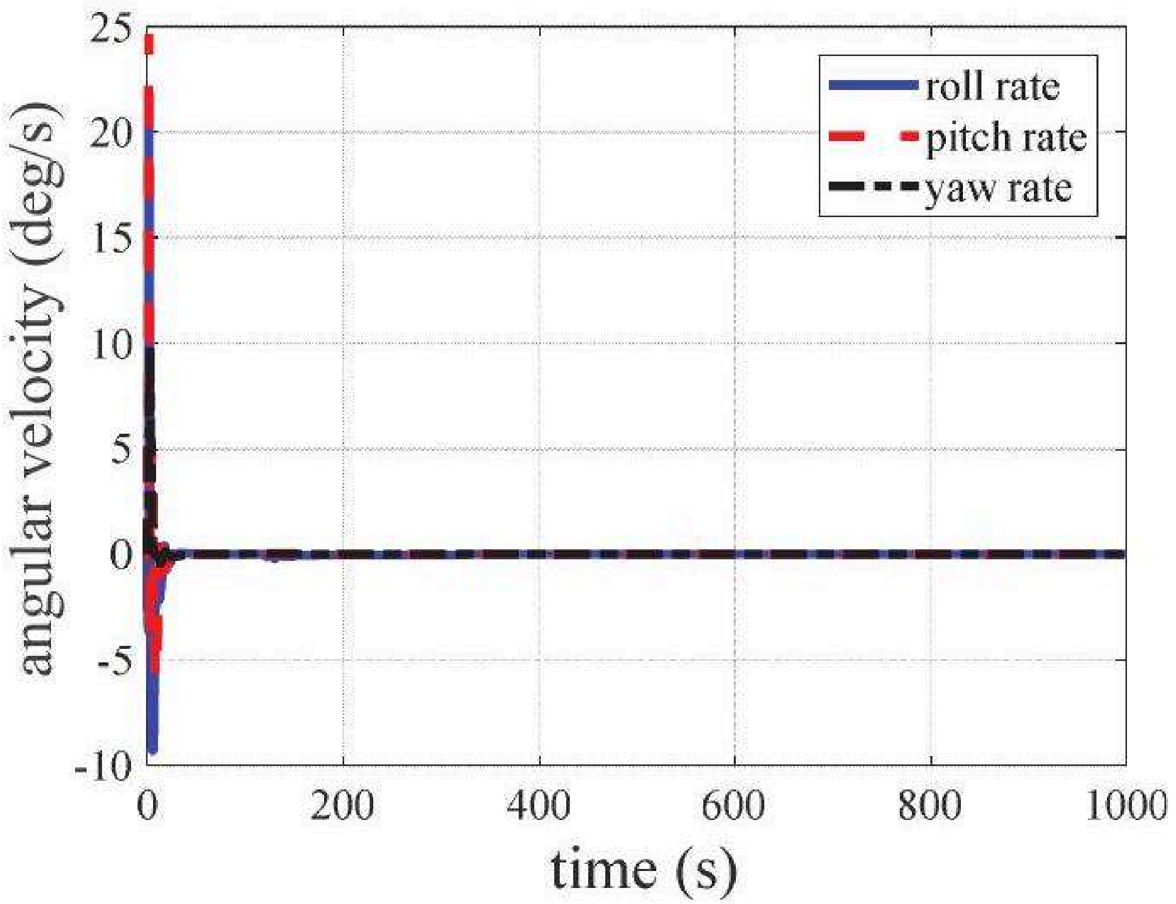

Figure 11: Fig. 11 Aircraft angular velocity under the reinforcement learning NN controller.

图 11：图 11 强化学习神经网络控制器下的飞机角速度。

In Fig. 12, the aircraft angle-of-attack and sideslip angle are presented. Different from the height and the velocity control, which reach the commanded value exactly. It is shown that the aircraft still maintains certain sideslip of -6.21° after entering the stable flight. Figure 13 gives the aero-surfaces deflection and throttle control of the aircraft. Since the reward function does not consider the penalty on the control inputs, the control transition process of the

在图 12 中，展示了飞机的迎角和侧滑角。与精确达到指令值的高度和速度控制不同，图中显示飞机在进入稳定飞行后仍保持一定的侧滑角 -6.21° 。图 13 给出了飞机的气动舵面偏转和油门控制。由于奖励函数没有考虑对控制输入的惩罚，因此控制的过渡过程

fixed-wing aircraft is not satisfactory. However, except the drastic changes at the initial stage, the control outputs all reach the equilibrium value gradually. To sum up, under the control of NN controller, despite of the fixed sideslip and roll, the aircraft enters a stable cruise mode and the control objective is achieved.

固定翼飞机的性能并不令人满意。然而，除了初始阶段的剧烈变化外，控制输出都逐渐达到平衡值。综上所述，在神经网络控制器的控制下，尽管侧滑和滚转是固定的，飞机仍进入稳定的巡航模式，并实现了控制目标。

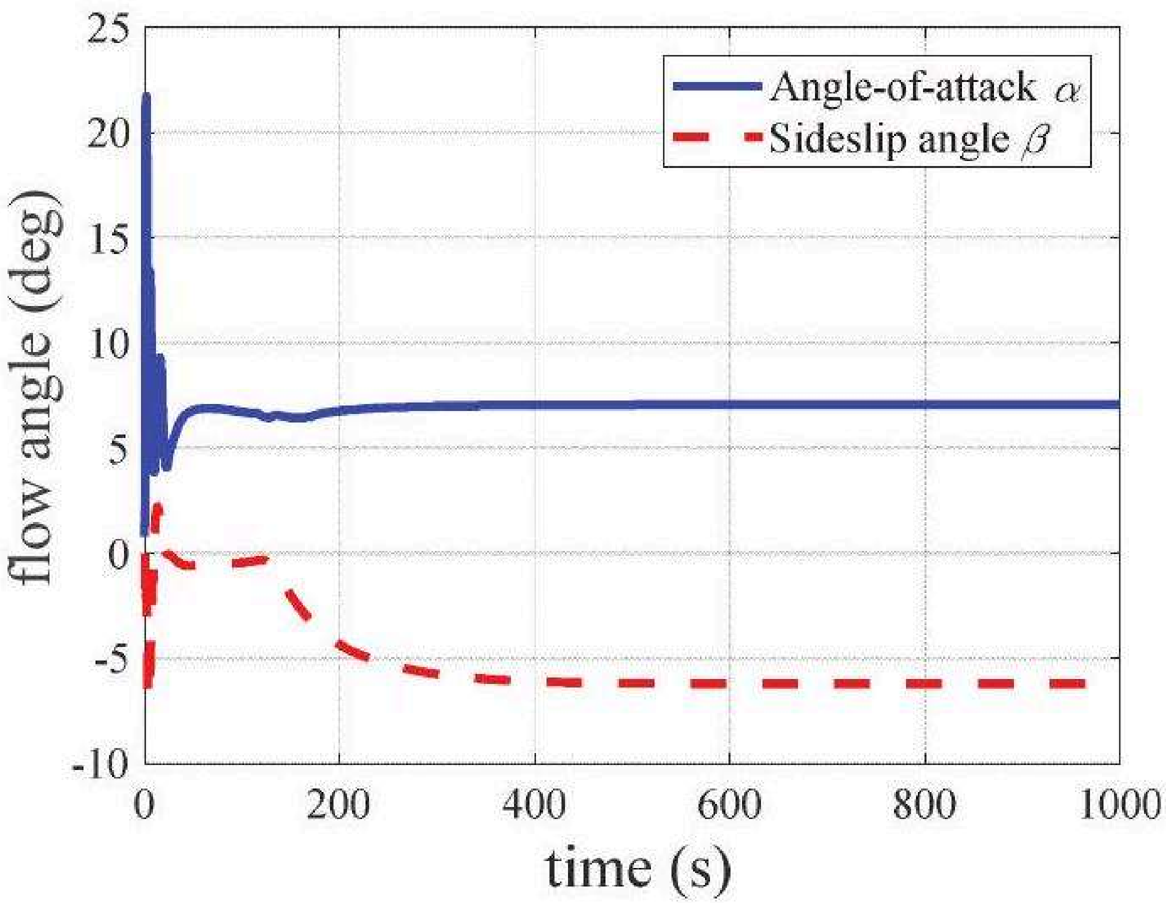

Figure 12: Fig. 12 The angle-of-attack and sideslip angle under the reinforcement learning NN controller.

图 12：强化学习神经网络控制器下的攻角和侧滑角。

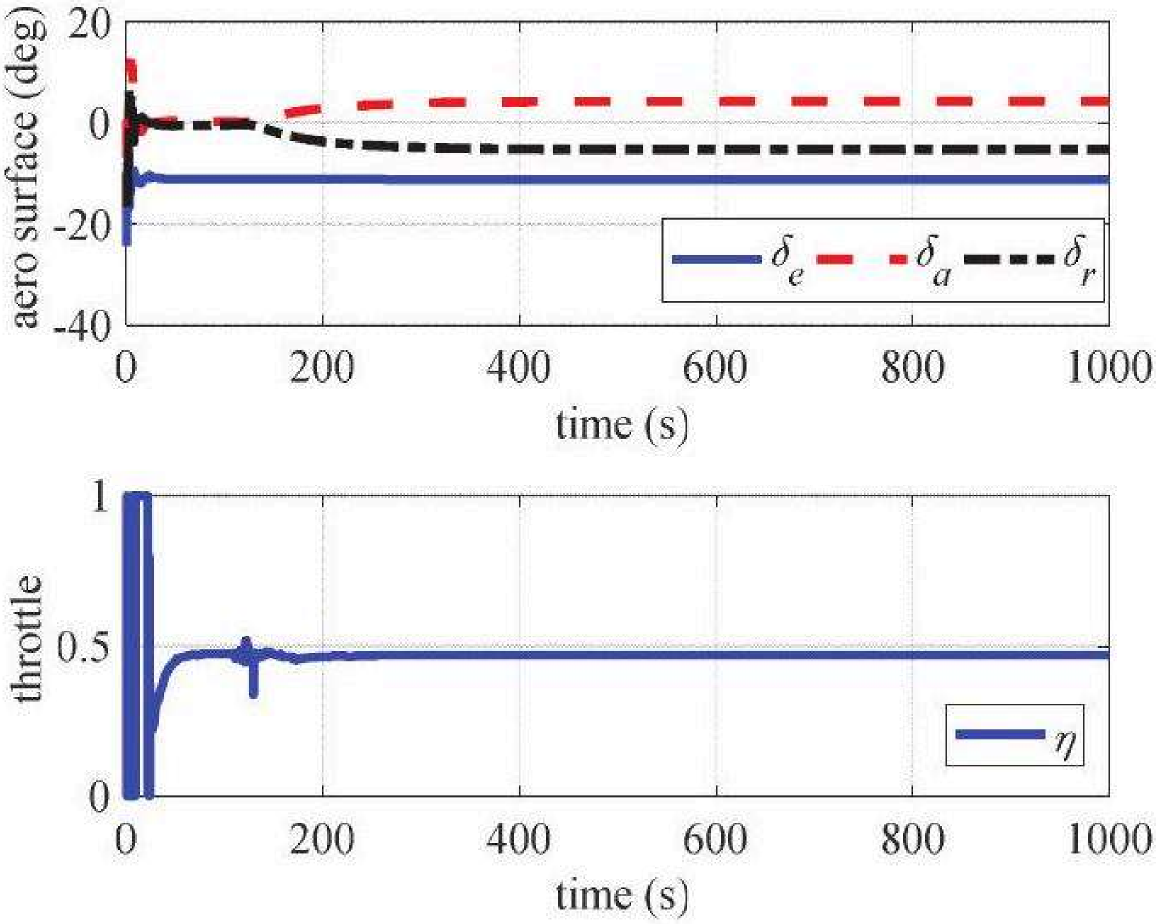

Figure 13: Fig. 13 The aero-surfaces and throttle control under the reinforcement learning NN controller.

图 13：图 13 强化学习神经网络控制器下的气动面和油门控制。

6 CONCLUSION

6 结论

Applying Neural Networks (NNs) for controller design is not novel. However, for the NN controllers studied before the Deep Reinforcement Learning (DRL), the NNs are mainly combined with traditional controllers to approximate the plant model, or used to control the plants of special inverse structure. In essence, those two manners approximate the system model, while with the DRL, the control law may be directly learned upon the generalization ability of NN models. Through this learning way, the end-to-end integrated control for the 6 degree-of-freedom flight of the fixed-wing aircraft is studied in this paper. Compared with the quadrotor, the implementation of intelligent control on the fixed-wing aircraft is more complicated since it needs to balance the aerodynamic force and the aerodynamic moment. To our encouragement, via the reinforcement learning, it is shown that the trained NN controller can learn the control law and achieve the control

将神经网络（NN）应用于控制器设计并非新颖。然而，在深度强化学习（DRL）之前的神经网络控制器研究中，神经网络主要与传统控制器结合以近似工厂模型，或用于控制具有特殊逆结构的工厂。从本质上讲，这两种方式都近似于系统模型，而通过 DRL，控制律可以直接在神经网络模型的泛化能力上学习。通过这种学习方式，本文研究了固定翼飞机六自由度飞行的端到端集成控制。与四旋翼飞机相比，固定翼飞机智能控制的实现更为复杂，因为它需要平衡气动力和气动力矩。令我们鼓舞的是，通过强化学习，结果表明训练后的神经网络控制器可以学习控制律并实现控制

objective. However, the control quality and generality of the NN controller still need to be improved, and these issues will be studied in the future.

目标。然而，神经网络控制器的控制质量和通用性仍有待提高，这些问题将在未来进行研究。

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